Nandini Pillai

Mechatronics Engineer

RESUME_1464

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Total Experience: 8 years

PROFESSIONAL SUMMARY

Mechatronics Engineer with 8 years of experience in product development, using CAD, SolidWorks, Arduino, ESP32. Experienced in electromechanical systems, integration tests and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: C++, Control Systems, CAD, ESP32, MATLAB, Motor Drivers, 3D Printing, PLC, Arduino, SolidWorks, Sensors, Python

Methods: Electromechanical systems, Integration tests, Product prototyping

PROFESSIONAL EXPERIENCE

Senior Mechatronics Engineer | HelioNex Labs | Jan 2023 - Present

- Validated 512 integration test cases across mechanical and electrical interfaces
- Improved prototype throughput by 18% through actuator timing optimization
- Raised measured actuator positioning consistency by 32% after calibration
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Mechatronics Engineer | Evergrid Technologies | Jan 2021 - Dec 2022

- Raised measured actuator positioning consistency by 19% after calibration
- Reduced assembly cost by 39% after redesigning electromechanical fixtures

Associate Mechatronics Engineer | NovaPeak Engineering | Jul 2018 - Dec 2020

- Validated 210 integration test cases across mechanical and electrical interfaces
- Improved prototype throughput by 30% through actuator timing optimization

PROJECTS

Smart Dispensing Unit | CAD, ESP32, C++, Motor Drivers

- Built controller firmware, mechanical enclosure and fault signals.
- Documented acceptance criteria and evaluated the design using integration tests and product prototyping.

Automated Sorting Prototype | CAD, Motor Drivers, MATLAB, C++

- Integrated optical sensors, actuator timing and conveyor control.
- Documented acceptance criteria and evaluated the design using integration tests and product prototyping.

EDUCATION

B.Tech in Mechatronics Engineering, Northbridge College of Technology, 2018

Relevant Methods: Electromechanical systems, Integration tests, Product prototyping

CERTIFICATIONS

Sensor Interfacing (2020)